

Intrinsic and extrinsic reinforced Triebel–Lizorkin spaces on a cube

Candidate full solution of Conjecture 4.22 in arXiv:1302.3751

Automated Mathematics Run `fa_banach_001`, lane 12

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Status. Candidate full solution, subject to human verification. For the entire parameter range in Conjecture 4.22,

$$F_{p,q}^{s,\text{rinf}}(Q) = F_{p,q}^{s,\text{rinf}}(Q)^*$$

with equivalent quasi-norms. More strongly, the proof constructs a bounded linear right inverse to restriction which preserves all reinforced face conditions simultaneously.

1 Source and exact conjecture

The source is B. Scharf, *Wavelets in function spaces on cellular domains* [1], arXiv:1302.3751. Let $Q = (0, 1)^n$, $n \geq 2$. For every open l -dimensional face Γ of Q , put $k = n - l$ and $d_\Gamma(x) = \text{dist}(x, \Gamma)$. A face is critical when

$$r_\Gamma = s - \frac{k}{p} \in \mathbb{N}_0.$$

For such a face the reinforced condition requires, for every multi-index α normal to Γ with $|\alpha| = r_\Gamma$,

$$d_\Gamma^{-k/p} D^\alpha f \in L_p(U_\Gamma), \quad U_\Gamma = \{x : d_\Gamma(x) < \varepsilon\}. \quad (1)$$

The extrinsic space $F_{p,q}^{s,\text{rinf}}(Q)$ consists of restrictions to Q of global $F_{p,q}^s(\mathbb{R}^n)$ functions satisfying (1) on both sides of every critical face. The intrinsic space $F_{p,q}^{s,\text{rinf}}(Q)^*$ asks for the same weighted estimates only on $U_\Gamma \cap Q$. Restriction immediately gives

$$F_{p,q}^{s,\text{rinf}}(Q) \hookrightarrow F_{p,q}^{s,\text{rinf}}(Q)^*.$$

Conjecture 4.22, on source PDF page 96, asserts that this embedding is an equality whenever

$$1 \leq p < \infty, \quad 0 < q < \infty, \quad s > \sigma_{p,q}.$$

The source explicitly proposes constructing an extension operator which preserves the reinforced conditions. Figure 1 records the conjecture and this proposed route.

We use the intrinsic quasi-norm

$$\|f\|_* = \|f\|_{F_{p,q}^s(Q)} + \sum_{\Gamma \text{ critical}} \sum_{\substack{\alpha \perp \Gamma \\ |\alpha| = r_\Gamma}} \left(\int_{U_\Gamma \cap Q} |D^\alpha f(x)|^p \frac{dx}{d_\Gamma(x)^k} \right)^{1/p}. \quad (2)$$

Changing the sufficiently small common value of ε changes this quasi-norm only by an equivalent one. Indeed, away from a smaller face collar the weight is bounded, while $D^\alpha f \in F_{p,q}^{s-|\alpha|}(Q) \subset L_p(Q)$ because $s - |\alpha| = k/p > 0$.

If $f \in F_{p,q}^{s,\text{rinf}}(Q)$, then there is by definition a continuation $\tilde{f}$ of f which belongs to $F_{p,q}^{s,\text{rinf}}(\mathbb{R}^n \setminus \Gamma)$. Then $\tilde{f}$ fulfils reinforce properties at Γ_l (if necessary) and surely also its restriction f fulfils interior reinforce properties at Γ_l . Hence we have

$$F_{p,q}^{s,\text{rinf}}(Q) \hookrightarrow F_{p,q}^{s,\text{rinf}}(Q)^*.$$

Thus the following statement would be nice to have.

Conjecture 4.22. Let Q be the unit cube in $\mathbb{R}^n$ for $n \geq 2$. Let

$$1 \leq p < \infty, \quad 0 < q < \infty \quad \text{and} \quad s > \sigma_{p,q}.$$

Then

$$F_{p,q}^{s,\text{rinf}}(Q) = F_{p,q}^{s,\text{rinf}}(Q)^*.$$

Remark 4.23. One idea to proof this conjecture is to show that $F_{p,q}^{s,\text{rinf}}(Q)^*$ admits the same decomposition as $F_{p,q}^{s,\text{rinf}}(Q)$ from Theorem 4.12, refined using the all-dimensional extension operator getting

$$F_{p,q}^{s,\text{rinf}}(Q) = F_{p,q}^{s,\text{rloc}}(Q) \times \text{Ext}_{\Gamma}^{\bar{r},u} \prod_{l=0}^{n-1} \prod_{\substack{\alpha \in \mathbb{N}_l^n \\ |\alpha| \leq r^l}} F_{p,p}^{s - \frac{n-l}{p} - |\alpha|, \text{rloc}}(\Gamma_l)$$

in (4.5).

A second possibility would be the following: Construct or use an existing (linear, bounded) extension operator

$$\text{Ext}_Q : F_{p,q}^s(Q) \rightarrow F_{p,q}^s(\mathbb{R}^n)$$

with $\text{Ext}_Q f|_Q = f$ and show that $\text{Ext}_Q f$ fulfils $R_l^{r^l,p}$ if f fulfils $R_l^{r^l,p,*}$ (with a suitable norm estimate). This would proof the conjecture.

Figure 1: Conjecture 4.22 and its extension-operator route on page 96 of the arXiv source [1].

2 Proof intuition

The cube admits an extension which, in a small collar, is a finite linear combination of coordinate reflections and dilations back into Q . A finite common Seeley–Hestenes reflection can be chosen bounded on the one fixed space $F_{p,q}^s$; no infinite reflection series is needed. Tensoring the construction over the coordinates gives, on every collar cell,

$$D^\alpha E_Q f(x) = \sum_A c_{A,\alpha} (D^\alpha f)(T_A x),$$

where each T_A is diagonal affine and maps the cell into Q .

The decisive geometric fact is one-sided in exactly the useful direction. If x is close to an open face Γ , then

$$d_\Gamma(T_A x) \leq C, d_\Gamma(x).$$

Normal distances are merely multiplied by one of finitely many slopes. Tangential coordinates which lie outside the open face interval are reflected into $(0, 1)$, so their contribution to the distance disappears. Consequently $d_\Gamma(x)^{-k} \leq C^k d_\Gamma(T_A x)^{-k}$. A change of variables therefore bounds the full

two-sided weighted norm of the extension by the one-sided intrinsic norm. The same finite operator works for all faces, dimensions, and critical derivative orders at once.

3 A finite reflection extension on the cube

We isolate the standard extension input in the form actually used below.

Lemma 1 (Fixed-parameter finite reflection extension). *Fix $n \geq 1$, $0 < p < \infty$, $0 < q < \infty$, and $s \in \mathbb{R}$. There is a bounded linear right inverse*

$$E_Q : F_{p,q}^s(Q) \longrightarrow F_{p,q}^s(\mathbb{R}^n)$$

to restriction with the following additional property. There are $\rho > 0$, a finite set of nonzero coefficients, and a finite set of positive slopes such that, on each cell of the ρ -collar of $\bar{Q}$, $E_Q f$ is a finite sum

$$E_Q f(x) = \sum_A c_A f(T_A x). \quad (3)$$

Every T_A is a diagonal affine map from that cell into Q ; in each coordinate it is the identity or a reflection/dilation about 0 or 1. The operator may be chosen compactly supported and its cutoff is identically one on the whole ρ -collar.

Proof. We recall why the usual common reflection extension has precisely this form. The quantitative common-extension theorem of Lu and Yao [2, Theorem 23] states that, after choosing sufficiently large integers m_1, m_2 , a half-space reflection

$$Rh(x', t) = \sum_{\nu=1}^N a_\nu h(x', -b_\nu t), \quad t < 0, \quad b_\nu > 0, \quad (4)$$

is bounded from $F_{p,q}^s(\mathbb{R}_+^n)$ to $F_{p,q}^s(\mathbb{R}^n)$ provided its finite coefficients satisfy

$$\sum_{\nu=1}^N a_\nu (-b_\nu)^j = 1 \quad (-m_1 \leq j \leq m_2).$$

Finite families satisfying as many of these moments as desired are given explicitly in [2, (29)–(30)]. We choose the moment range larger than every derivative order used below.

Translate and reflect (4) to treat both endpoints of $(0, 1)$. A smooth partition of unity consisting of a left endpoint piece, an interior piece, and a right endpoint piece localizes the half-space theorem; the interior piece is extended by zero. The endpoint cutoffs are constant on smaller collars. This produces a bounded interval extension which, on those collars, is exactly the appropriate finite reflection formula. To spell out the product step, set $D_j = \mathbb{R}^j \times (0, 1)^{n-j}$ for $0 \leq j \leq n$. Extend from D_{j-1} to D_j by applying the interval formula in coordinate j . If g is any global representative of an element of $F_{p,q}^s(D_{j-1})$, the localized half-space theorem gives a global representative of this new function with norm at most $C\|g\|_{F_{p,q}^s(\mathbb{R}^n)}$. Taking the infimum over g proves boundedness $F_{p,q}^s(D_{j-1}) \rightarrow F_{p,q}^s(D_j)$. Composing these n maps gives the asserted cube extension.

Expanding the finite tensor product yields (3). Finally, multiply by a smooth compactly supported function which is one on a still smaller collar of $\bar{Q}$. Smooth multiplication is bounded on $F_{p,q}^s$, and all claimed properties remain valid after decreasing ρ . $\square$

For a face it is useful to make its distance completely explicit. Write

$$\Gamma = \Gamma_{I,\epsilon} = \{x \in \mathbb{R}^n : x_i = \epsilon_i \ (i \in I), \ 0 < x_j < 1 \ (j \notin I)\}, \quad (5)$$

where $\epsilon_i \in \{0, 1\}$ and $|I| = k$. If $\delta(t) = \text{dist}(t, (0, 1))$, then

$$d_\Gamma(x)^2 = \sum_{i \in I} |x_i - \epsilon_i|^2 + \sum_{j \notin I} \delta(x_j)^2. \quad (6)$$

Lemma 2 (Every face weight is reflection-stable). *After decreasing ρ in Lemma 1, there is $B \geq 1$ such that for every open face Γ , every collar cell meeting $\{\delta_\Gamma < \rho\}$, every affine term T_A on that cell, and every point x in the intersection,*

$$d_\Gamma(T_A x) \leq B d_\Gamma(x). \quad (7)$$

The constant is independent of the face and of the cell.

Proof. Take $\rho < 1/4$ and smaller, if necessary, so that every reflected point in the construction lies in Q . For $i \in I$, closeness to Γ forces x_i to be close to the specified endpoint ϵ_i , not the opposite endpoint. The corresponding component of T_A is either the identity or a reflection/dilation about ϵ_i , and hence

$$|(T_A x)_i - \epsilon_i| \leq B |x_i - \epsilon_i|$$

for the maximum B of the finitely many slopes. For $j \notin I$, $(T_A x)_j \in (0, 1)$, so $\delta((T_A x)_j) = 0$. Formula (6) proves (7). $\square$

4 The conjecture

Theorem 3 (Conjecture 4.22). *Let $Q = (0, 1)^n$, $n \geq 2$, and suppose*

$$1 \leq p < \infty, \quad 0 < q < \infty, \quad s > \sigma_{p,q}.$$

Then

$$F_{p,q}^{s,\text{rinf}}(Q) = F_{p,q}^{s,\text{rinf}}(Q)^*$$

with equivalent quasi-norms. In fact the operator E_Q in Lemma 1 restricts to a bounded linear map

$$E_Q : F_{p,q}^{s,\text{rinf}}(Q)^* \longrightarrow F_{p,q}^{s,\text{rinf}}(\mathbb{R}^n \setminus \partial Q)$$

and remains a right inverse to restriction.

Proof. Let $f \in F_{p,q}^{s,\text{rinf}}(Q)^*$ and set $F = E_Q f$. Its ordinary $F_{p,q}^s(\mathbb{R}^n)$ norm is at most a constant times $\|f\|_{F_{p,q}^s(Q)}$ by Lemma 1. Fix a critical face $\Gamma = \Gamma_{I,\epsilon}$, write $k = |I|$, and let $r = s - k/p \in \mathbb{N}_0$. Fix a normal multi-index α , so $\alpha_j = 0$ for $j \notin I$ and $|\alpha| = r$.

Choose the reinforced collar inside the region where the cutoff from Lemma 1 is one. On the interior of each collar cell, differentiating the finite affine formula gives

$$D^\alpha F(x) = \sum_A c_{A,\alpha} (D^\alpha f)(T_A x). \quad (8)$$

This also holds distributionally across cell boundaries; alternatively, those boundaries have measure zero for the weighted L_p estimate. The moment order in Lemma 1 was chosen larger than r .

For a collar cell C , the finite-sum inequality (using $p \geq 1$), followed by a change of variables $y = T_A x$, gives

$$\begin{aligned} & \int_{C \cap U_\Gamma} |D^\alpha F(x)|^p \frac{dx}{d_\Gamma(x)^k} \\ & \leq C \sum_A \int_{C \cap U_\Gamma} |D^\alpha f(T_A x)|^p \frac{dx}{d_\Gamma(x)^k} \\ & \leq C \sum_A \int_{T_A(C \cap U_\Gamma)} |D^\alpha f(y)|^p \frac{dy}{d_\Gamma(y)^k}. \end{aligned}$$

For the second inequality we used (7) in the form $d_\Gamma(x)^{-k} \leq B^k d_\Gamma(T_A x)^{-k}$, together with the uniformly bounded Jacobians of the finitely many affine maps. Moreover $T_A(C \cap U_\Gamma)$ lies in $Q \cap \{d_\Gamma < B\varepsilon\}$.

There are only finitely many cells and affine terms. Summing, and using the independence of the intrinsic condition from the precise small collar width, yields

$$\left(\int_{U_\Gamma} |D^\alpha F(x)|^p \frac{dx}{d_\Gamma(x)^k} \right)^{1/p} \leq C \left(\int_{Q \cap U'_\Gamma} |D^\alpha f(y)|^p \frac{dy}{d_\Gamma(y)^k} \right)^{1/p}. \quad (9)$$

The construction and estimate apply simultaneously to every critical face and every required normal multi-index. Thus

$$\|E_Q f\|_{F_{p,q}^{s,\text{rinf}}(\mathbb{R}^n \setminus \partial Q)} \leq C \|f\|_*.$$

Because $E_Q f|_Q = f$, this proves $F_{p,q}^{s,\text{rinf}}(Q)^* \hookrightarrow F_{p,q}^{s,\text{rinf}}(Q)$. The reverse embedding is the immediate restriction observation from the source, and the displayed estimates give equivalence of quasi-norms. $\square$

Corollary 4 (The intrinsic Sobolev example). *For the unit square $Q \subset \mathbb{R}^2$,*

$$W_2^{1,\text{rinf}}(Q)^* = \left\{ f \in W^{1,2}(Q) : \int_Q \frac{|f(x)|^2}{\text{dist}(x, \Gamma_0)^2} dx < \infty \right\} = F_{2,2}^{1,\text{rinf}}(Q),$$

where Γ_0 is the set of four vertices, with equivalent norms. Consequently the intrinsic space has an oscillating u -Riesz basis for every integer $u > 1$.

Proof. For $n = 2$, $s = 1$, and $p = q = 2$, the only critical stratum has $l = 0$, $k = 2$, and $r = 0$, so (1) is exactly the displayed corner condition. Theorem 3 identifies the intrinsic and extrinsic spaces. The source's wavelet theorem for the latter therefore transfers to the former. $\square$

5 Verification, scope, and novelty status

The proof is analytic; no numerical experiment is needed. The following checks are explicit in the argument.

- The external functional-space input is precisely the finite common-extension case of Lu–Yao Theorem 23, not their infinite universal reflection series. Finiteness is what makes the weighted argument immediate.
- Formula (6) includes the tangential endpoint contribution to the distance from an *open* face. Under reflection that contribution becomes zero; it is not silently discarded.

- The cutoff is one on all reinforced collars, so no derivative of the cutoff appears in (8).
- The restriction $p \geq 1$ is used exactly in the finite-sum L_p estimate. The proof covers every $0 < q < \infty$ in the conjecture.

The result is for an axis-parallel cube. Extending the same proof to a general polyhedron would require a separate construction compatible with intersecting non-orthogonal faces and is not claimed here.

Targeted searches through 9 August 2026 for the arXiv id, the exact intrinsic/extrinsic reinforced-space notation, Conjecture 4.22, and the extension problem found the source thesis and related reinforced-space work, but no paper stating this equality. Lu and Yao prove the reflection extension theorem used here, but do not discuss reinforced face weights or Scharf’s conjecture. Novelty is therefore plausible, not certified.

References

- [1] B. Scharf, *Wavelets in function spaces on cellular domains*, arXiv:1302.3751 (2013).
- [2] H. Lu and L. Yao, *On Seeley-type universal extension operators for the upper half space*, arXiv:2211.15567v2 (2024), especially Theorem 23 and equations (29)–(30).